

Active disturbance rejection control of pressurized water reactor[☆]

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ARTICLE INFO

Keywords:

Pressurized water reactor
Active disturbance rejection control
Extended state observer
Measurement noise

ABSTRACT

Control design for pressurized water reactor (PWR) is difficult due to associated non-linearity, modelling uncertainties and time-varying system parameters. Extended state observer (ESO) based active disturbance rejection control (ADRC) presents a simple and robust solution which is almost model free and has few tuning parameters. However, conventional ESO suffers from noise over-amplification in the obtained estimates due to high-gain construction which in turn degrades the noise sensitivity of the closed-loop system and limits the achievable dynamic performance in practical scenarios. To overcome this problem, two recent techniques namely cascade ESO (CESO) and low-power higher-order ESO (LHESO) are implemented for control of PWR. Simulation analysis is conducted in MATLAB to illustrate the performance improvement obtained over conventional ESO based ADRC, particularly in case of time-varying disturbances. Extensive simulation analysis is also conducted to investigate robustness towards parametric uncertainties. The study also presents a comparison of conventional ESO, CESO and LHESO to highlight their advantages as well as limitations which in turn would help the users in selecting the appropriate scheme for their particular use-case.

1. Introduction

The world is grappling with increasing demand for reliable, sustainable, and clean energy. There are several sources of energy including renewables, carbon-based or fossil fuels, and nuclear energy. Carbon-based energy sources are responsible for the increase in the global average temperature due to greenhouse gases. In addition, the carbon-based energy sources cause pollution with concomitant effects on human health and the environment. Consequently, efforts to reduce pollution and global temperatures to avoid catastrophic events include substituting fossil fuels with clean energy sources. Renewable energy sources including solar, and wind energy have been identified as replacements for fossil fuels. However, they are incapable of providing bulk energy as fossil fuels. For instance, they are weather dependent and require a large expanse of land. Instead, nuclear energy can be used as an adequate and effective substitute for fossil fuels. Technological advancements have enabled the application of nuclear power plants in the load-following mode, thereby making nuclear power plants operable as base load, or peak load plants. Recently, studies have been conducted on hybrid power systems that combine nuclear energy and renewable energy to complement each other. Nonetheless, safe and

reliable application of nuclear energy requires efficient control systems to avoid accidents that can lead to fuel melting and the subsequent release of radioactive materials into the environment. Researchers and practitioners have considered several control techniques for nuclear reactor core power control systems to improve the system performance and availability. Among the techniques that have been applied are the feedback and optimal control techniques. A state feedback controller was designed based on a linear quadratic regulator (LQR) control approach to design a core control system for a CANDU heavy water reactor (Xia et al., 2014). Li applied the optimal control approach of linear quadratic gaussian/ loop transfer recovery (LQG/LTR) to regulate the core power of a PWR in load-following mode using a multi-model approach (Li and Zhao, 2013a). The state feedback-assisted classical control technique (SFACC) is widely applied in the nuclear reactor domain. State feedback which uses a classical differential lag compensator assisted by linear quadratic gaussian with loop transfer recovery (LQG/LTR) was used to control the core power of a PWR (Wan and Wang, 2020). Adaptive state feedback based on Model reference adaptive control that uses a normalized negative gradient algorithm was also used to design a nuclear reactor core control system (Reddy

[☆] The work of H. Ahmed is funded through the Sêr Cymru II 80761-BU-103 project by Welsh European Funding Office (WEFO) under the European Regional Development Fund (ERDF). This work was supported in part by the Royal Society Short Industry Fellowship under grant SIF\R1\221035.

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<https://doi.org/10.1016/j.anucene.2023.109845>

Received 29 April 2022; Received in revised form 8 February 2023; Accepted 2 April 2023

Available online 6 April 2023

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et al., 2020). To further enhance the optimal state feedback control, an LQG, PID and an optimization Improved Adaptive Genetic Algorithm (IAGA) were combined to synthesize a new control paradigm for the core power control of a nuclear reactor (Li and Zhao, 2013b). Nonetheless, Optimal state feedback control performance degrades with time in an uncertain environment such as a nuclear reactor, and it is not robust against external disturbances because the control system design does not consider uncertainties in the system (Surjagade et al., 2020). In addition, none of the control paradigms considered external disturbances in their design. Thus, hybrid control techniques were designed to improve the robustness of the optimal control techniques against external disturbances. The hybrid control paradigm combines LQG/LTR with sliding mode control (Surjagade et al., 2020; Abdurraheem and Korolev, 2021), and Linear Quadratic Integrator, Linear Quadratic Gaussian and Loop Transfer Recovery (LQGI/LTR) (Vajpayee et al., 2021). The primary difference between LQG/LTR and LQGI/LTR is that LQGI/LTR is a hybrid control technique that integrates linear quadratic integrator, linear quadratic gaussian and loop transfer recovery unlike LQG/LTR that is a technique for improving robustness of LQG through loop transfer recovery (Vajpayee et al., 2021). Other control techniques that have been used to address the control challenges in the nuclear domain include model predictive control. A model predictive control technique was used in the load-following operation of a nuclear reactor (Wang et al., 2017; Kim et al., 2014). Model predictive control possesses superior performance for constrained systems such as nuclear reactor. Nevertheless, it is computationally complex as it requires solving optimization problem online when both state and control input constraints are considered. In addition, external disturbance and/or changes in plant parameters deteriorate the control performance, which may require additional parameter and/or disturbance estimator/observer (Zhang et al., 2021; Yang et al., 2015). Intelligent control has also been explored for nuclear reactors control. Intelligent control techniques for nuclear reactors control include fuzzy logic control (Luan et al., 2021), and artificial neural network techniques such as gradient descent algorithm (Ejigu and Liu, 2022). However, the intelligent control technique is an emerging technology that will take time to be applied in risky technologies such as nuclear power plants. Moreover, the disturbance rejection capability of the control system has not been fully examined.

Furthermore, a nuclear power plant is a time-varying, nonlinear, and complex system. Therefore, a robust nonlinear control technique that is insensitive to parametric variations and external disturbances is vital for controlling the system. Sliding mode control is a robust nonlinear control method that has been extensively studied for nuclear power plant control (Ansarifar et al., 2016, 2018). However, chattering is the main drawback of this control technique. Recently, researchers have conducted studies to address the challenges of sliding mode control (Abdurraheem et al., 2021, 2022). It should be noted that all approaches to reject disturbances and uncertainties in all the control systems discussed thus far are passive disturbance attenuation or rejection. Thus, a new control paradigm to address disturbances in nuclear power plant control systems is required for system safety, efficiency, and reliability. This new approach actively rejects disturbances in complex systems such as nuclear power plants. Recently, several studies have used the active disturbance rejection technique in nuclear reactor control loops owing to its simplicity and effective solution to nuclear power plant control challenges. Besides, it is simple to implement and tune, and requires minimal plant information in terms of the relative degree and input gain (Sun et al., 2021; Hui et al., 2020; Liu et al., 2020; Hui and Yuan, 2022b). However, most of the studies on active disturbance rejection control techniques do not consider the effect of measurement noise which is ubiquitous in practical systems and degrades the quality of estimates obtained via extended state observers (ESO) and severely limits the achievable closed-loop performance. Motivated by the gap observed in this area, we apply two recently proposed noise suppressing ESOs namely low-power higher-order ESO

Table 1

Nomenclature used for modelling of PWR.

n_r	- Neutron density relative to initial equilibrium point
n_{r0}	- Initial equilibrium neutron density
c	- Precursor density
c_0	- Equilibrium precursor density at rated power
λ	- Effective precursor radioactive decay constant
ρ	- Total reactivity
ρ_r	- Reactivity induced by the control mechanism
Λ	- Effective prompt neutron life time
β	- Total delayed neutron fraction
T_e	- Temperature of water entering the reactor
z_r	- Control input, control rod speed in units of fraction of core length per second
G_r	- Total reactivity worth of control rod
α_f	- Fuel temperature reactivity coefficient
α_c	- Coolant temperature reactivity coefficient
T_f	- Average reactor fuel temperature
T_{f0}	- Initial equilibrium average reactor fuel temperature
T_l	- Temperature of water leaving the reactor
T_{l0}	- Initial equilibrium (steady state) temperature of water leaving the reactor
μ_f	- Total heat capacity of fuel
μ_c	- Total heat capacity of reactor coolant
Ω	- Heat transfer coefficient between fuel and coolant
M	- Mass flow rate multiplied by heat capacity of coolant
f_f	- Fraction of reactor power deposited in the fuel
P_0	- Rated Thermal power

(LHESO) (Ahmad and Ali, 2021a) and cascade ESO (CESO) (Łakomy and Madonski, 2021; Łakomy et al., 2022) for control of PWR under the effect of non-negligible sensor noise. Through extensive simulation studies, we illustrate that the noise suppressing ESO based ADRC techniques offer improved resilience towards high-frequency measurement noise along with better closed loop performance in terms of reference tracking as well as disturbance attenuation. In addition to the nominal operating condition, eight different perturbed scenarios with positive as well as negative deviations of critical PWR parameters are considered to analyse the robustness of the developed schemes.

A preliminary version of this article is published in Ahmad et al. (2022). However, different from Ahmad et al. (2022), this article also includes the following results and analyses to highlight the effectiveness of noise suppressing ESOs: (a) Convergence analysis of the LHESO having two state augmentations and level two CESO designed for PWR considering the effect of disturbances as well as measurement noise. (b) A more detailed comparison of ESO, CESO as well as LHESO in terms of noise suppression and disturbance attenuation to highlight their structural properties and advantages as well as limitations. (c) Analysis of tracking and disturbance attenuation under ramp type signals for nominal as well as perturbed scenarios. (d) Performance comparison of the closed-loop system under actuator constraints. More simulation scenarios as well as in-depth analysis and discussions included in the present work is expected to aid the control practitioners in selecting the appropriate scheme for individual use-cases.

2. Modelling of pressurized water reactor

Generally, the Boltzmann transport equation captures time-dependent flux distribution in a nuclear reactor. However, exact solutions of the equation can be obtained for a few rare scenarios. Nevertheless, the approximate neutron diffusion equation where the energy variable is discretized and intergroup transfer cross sections represent the slowing-down process can adequately be used instead of the transport equation (Shimjith et al., 2010). In addition, the control system design requires simple systems of first-order differential equations. Therefore, the neutron diffusion equation, including delayed neutrons, was used for the derivation of the reactor Point kinetics equation, which is a set of first-order equations. The point kinetics equation is a fundamental equation that describes the time-dependence

of reactor core multiplication. It can be used to predict the behaviour of the neutron population in a reactor (Duderstadt and Hamilton, 1976). The detailed derivation of the equation is given by Duderstadt and Hamilton (1976), Hetrick (1971).

The Point kinetics model, which accounts for the neutronic model is coupled with other models including the thermal-hydraulics and reactivity models. The thermal-hydraulics model account for the temperature reactivity feedback owing to the doppler effect and coolant. The thermal-hydraulic model is based on a singly lumped model of the fuel and coolant temperatures. Finally, the reactivity model accounts for the total reactivity including the external reactivity of the control rods. Therefore, the model comprises three main parts: the neutronic model, thermal-hydraulics model, and reactivity model. Therefore, the reactivity model comprises the reactivity owing to control rod, the coolant, and fuel temperature. Following the approach used in Luan et al. (2021), Ejigu and Liu (2022), Hui et al. (2020), Liu et al. (2020), Hui and Yuan (2022a), the complete model for the nuclear reactor is expressed as follows:

$$\frac{dn_r}{dt} = \frac{\rho - \beta}{\Lambda} n_r + \frac{\beta}{\Lambda} c_r \quad (1)$$

$$\frac{dc_r}{dt} = \lambda(n_r - c_r) \quad (2)$$

$$\frac{dT_f}{dt} = \frac{f_f P_0}{\mu_f} n_r - \frac{\Omega}{\mu_f} T_f + \frac{\Omega}{2\mu_f} (T_l + T_e) \quad (3)$$

$$\frac{dT_l}{dt} = \frac{(1 - f_f) P_0}{\mu_c} n_r + \frac{\Omega}{\mu_c} T_f - \frac{(2M + \Omega)}{2\mu_c} T_l + \frac{(2M - \Omega)}{2\mu_c} T_e \quad (4)$$

$$\frac{d\rho_r}{dt} = G_r z_r \quad (5)$$

$$\rho = \rho_r + \alpha_f (T_f - T_{f0}) + 0.5\alpha_c (T_l - T_{l0}) + 0.5\alpha_e (T_e - T_{e0}) \quad (6)$$

$$P = P_0 n_r \quad (7)$$

where

$$\alpha_f(n_r) = (n_r - 4.24) \times 10^{-5} \frac{\delta k}{k} / ^\circ\text{C}$$

$$\alpha_c(n_r) = -(4n_r + 17.3) \times 10^{-5} \frac{\delta k}{k} / ^\circ\text{C}$$

$$\mu_c(n_r) = \left(\frac{160}{9} n_r + 54.022 \right) MW \cdot s / ^\circ\text{C} \quad (8)$$

$$\Omega(n_r) = \left(\frac{5}{3} n_r + 4.933 \right) MW / ^\circ\text{C}$$

$$M(n_r) = (28n_r + 74) MW / ^\circ\text{C}.$$

Nomenclature used for the modelling the PWR is given in Table 1.

3. Active disturbance rejection control (ADRC)

Dynamic model of a PWR expressed in (1)–(8) is highly nonlinear and has time-varying parameters. Furthermore, using the entire higher-order system dynamics may lead to a complex control structure that is difficult to design and implement. Active disturbance rejection control (ADRC) (Han, 2009; Ahmad and Ali, 2019) presents a simple and intuitive solution to this problem by considering the nominal model as a cascaded chain of integrators where the system order relies on the relative degree of the actual system and only approximate information about the input gain of the system is required. In doing so, ADRC expands the notion of disturbance by considering only the required minimal system information for controller design and categorizing any deviation from the assumed nominal model as total disturbance (Ahmad et al., 2019).

Rewriting the neutron dynamics as

$$\frac{dn_r}{dt} = -\frac{\beta}{\Lambda} (n_r - c_r) + \frac{n_r}{\Lambda} \left[\int_0^t G_r z_r d\tau + \alpha_f \Delta T_f + 0.5\alpha_c (\Delta T_l + \Delta T_e) \right],$$

using (1) and (6), it can be observed that the relative degree of the PWR dynamics is two as the control input z_r appears in the second derivative of the preceding neutron dynamics. It is therefore possible to rewrite the 5th order dynamics of PWR using a second order integral chain form given by

$$\frac{d^2 n_r}{dt^2} = b_0 z_r + f \quad (9)$$

where $b_0(n_r) = G_r n_r / \Lambda$ is the input gain and f is the total disturbance expressed as

$$f = \frac{G_r(n_r - n_{r0})}{\Lambda} z_r + \frac{n_r}{\Lambda} [\dot{\alpha}_f \Delta T_f + \alpha_f \Delta \dot{T}_f + 0.5\dot{\alpha}_c (\Delta T_l + \Delta T_e) + 0.5\alpha_c (\Delta \dot{T}_l + \Delta \dot{T}_e)] - \frac{\beta}{\Lambda} (\dot{n}_r - \dot{c}_r) + \frac{\dot{n}_r}{\Lambda} \left[\int_0^t G_r z_r d\tau + \alpha_f \Delta T_f + 0.5\alpha_c (\Delta T_l + \Delta T_e) \right]. \quad (11)$$

Considering the total disturbance f as a virtual state variable (x_3) that needs to be estimated along with the system states (x_1, x_2), the augmented state-space model is expressed as

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = x_3 + b_0 u \\ \dot{x}_3 = h \\ y = x_1 + v \end{cases} \quad (12)$$

where $x_1 = n_r$ is the output to be controlled, $u = z_r$ is the control input, $h = \dot{f}$ and v denotes the effect of measurement noise. The compact matrix form for (12) is given by

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + b_0 \mathbf{B}u + \mathbf{E}h \\ y &= \mathbf{C}\mathbf{x} + v, \end{aligned} \quad (13)$$

where $\mathbf{x} = [x_1, x_2, x_3]^T$ and the associated matrices are defined as

$$\mathbf{A} = \begin{bmatrix} 0 & I_2 \\ 0 & 0 \end{bmatrix}_{3 \times 3}, \quad \mathbf{B} = [0 \quad 1 \quad 0]_{1 \times 3}^T, \quad \mathbf{C} = [1 \quad 0 \quad 0]_{1 \times 3}, \quad \mathbf{E} = [0 \quad 1]_{1 \times 3}^T$$

Assuming that the state as well as disturbance estimates are available, a disturbance decoupling control law of the form

$$u = b_0^{-1} [-\hat{x}_3 + \bar{u}], \quad (14)$$

can be implemented to reduce the perturbed nominal model to a pure integrating form while a stabilizing nominal controller can be selected as

$$\bar{u} = k_1(r - \hat{x}_1) - k_2 \hat{x}_2 \quad (15)$$

to obtain desired closed-loop dynamics. The required estimates of x_i , $i = \{1, 2, 3\}$ are obtained via an extended state observer (ESO) designed for the augmented state-space model expressed in (13).

As highlighted in Ahmad and Ali (2021a), Łakomy and Madonski (2021), Łakomy et al. (2022), Ahmad and Ahmed (2022), conventional ESO suffers from noise amplification problem in the obtained estimates due to the high-gain construction. This in turn degrades the control quality and limits the closed-loop performance that can be achieved via ADRC. To address this problem, this paper investigates the implementation of noise suppressing ESOs in the context of PWR control for enhanced closed-loop performance. Design and implementation of noise suppressing ESOs are discussed in the subsequent sections.

4. Cascade extended state observer (CESO)

Cascade ESO (Łakomy and Madonski, 2021; Łakomy et al., 2022) operates by virtually splitting the total disturbance (f) into a predefined number of components ($f_i, i = \{1, \dots, N\}$), each lying in a particular frequency range. These components are then estimated via N individual

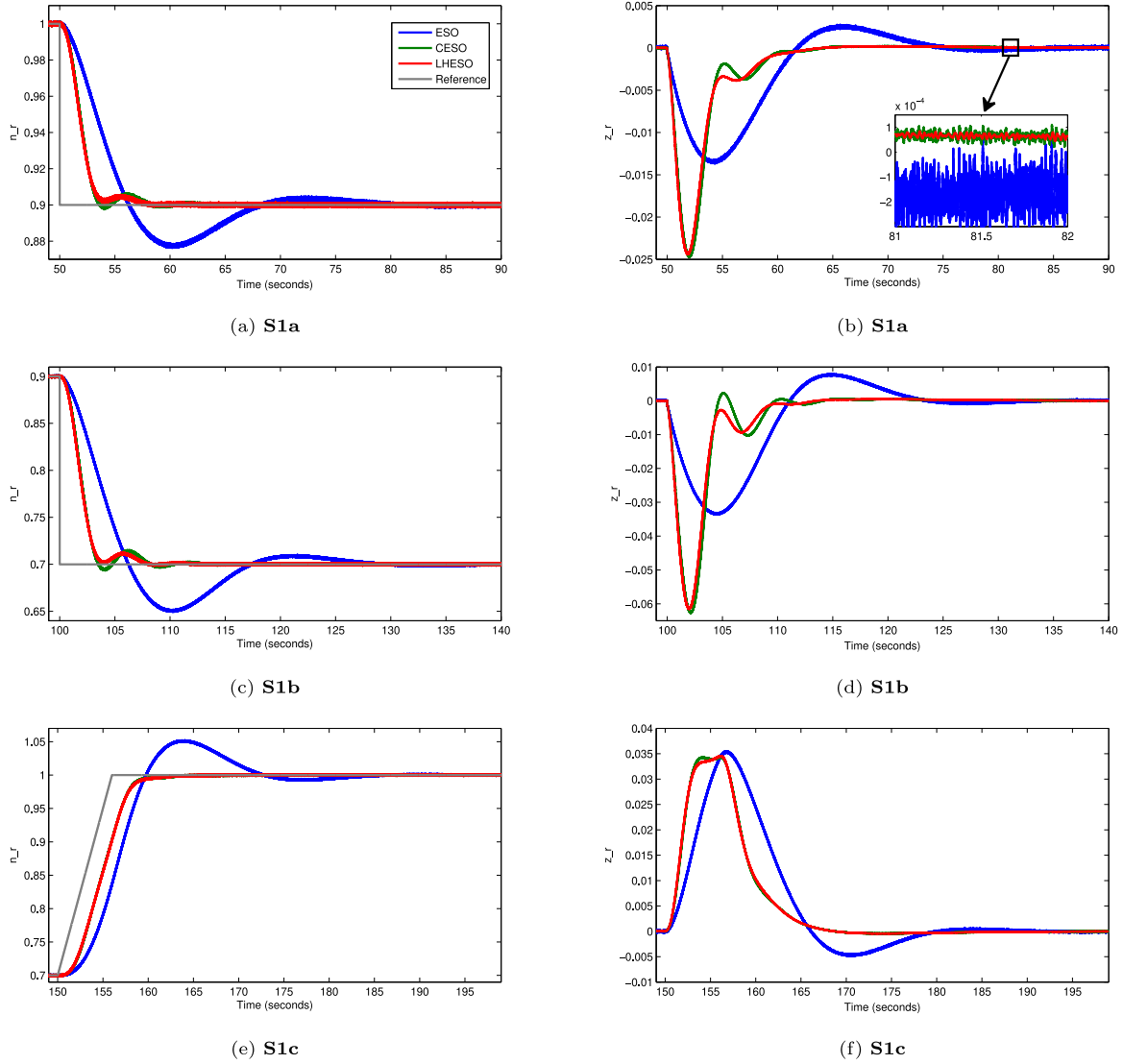


Fig. 1. Simulation results for *Scenario 1* (S1a, S1b, S1c) with variations in n_r (on the left) and applied control signal z_r (towards the right).

ESOs connected in cascade where the output of an ESO in each level acts as reference for the ESO in subsequent level.

The ESO in each level is a Luenberger type observer constructed for the augmented state-space model given in (13) where the expression for i th level ESO is given as follows:

$$\dot{\hat{z}}_i = \mathcal{A}\hat{z}_i + b_0Bu + \gamma_i(y_{i-1} - C\hat{z}_i) + \sum_{k=1}^{i-1} \Gamma\hat{z}_k, \quad (16)$$

where

$$\Gamma = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \hat{z}_i = [\hat{z}_{i,1}, \hat{z}_{i,2}, \hat{z}_{i,3}]^T, \gamma_i = [\gamma_{1,i}\omega_{oi}, \gamma_{2,i}\omega_{oi}^2, \gamma_{3,i}\omega_{oi}^3]^T,$$

$$y_i = C\hat{z}_i, i = \{1, \dots, N\}, y_0 = y.$$

In order to reduce the number of tuning parameters in the ESOs, the observer gains are selected such that all the poles are placed at $-\omega_{oi}$ as per bandwidth parameterization approach highlighted in Gao (2006), which results in $\gamma_{1,i} = 3, \gamma_{2,i} = 3, \gamma_{3,i} = 1$, for third order ESOs used in the present study. Furthermore, the observer bandwidths for individual ESOs are selected as $\omega_{oi} = \omega_o a^{i-1}$, $i = \{1, \dots, N\}$ where $a > 1$ is a tuning parameter.

The final set of estimates of the system states as well as total disturbance are obtained as

$$\hat{x} = [\hat{x}_1, \hat{x}_2, \hat{x}_3]^T = \hat{z}_N + EE^T \sum_{i=1}^{N-1} \hat{z}_i, \quad (17)$$

where the state estimates of system states i.e. $\hat{x}_1, \hat{x}_2$ are obtained from the N th level ESO while estimate of total disturbance is a sum of disturbance estimates obtained from each ESO.

We define a combined estimation error variable as $e_z := [e_{z_1}^T, e_{z_2}^T, \dots, e_{z_N}^T]^T$ where

$$e_{z_i} := x - \hat{x}_i, \hat{x}_i = \hat{z}_i + EE^T \sum_{j=1}^{i-1} \hat{z}_j, i = \{1, 2, \dots, N\}, \quad (18)$$

following the approach introduced in Lakomy et al. (2022). Considering a two level CESO which is used in the present study, the combined estimation error dynamics is given as

$$\dot{e}_z = \mathcal{A}_z e_z + E_z h - \gamma_z v \quad (19)$$

where

$$\mathcal{A}_z = \begin{bmatrix} \mathcal{A} - \gamma_1 C & \mathbf{0}_{3 \times 3} \\ \gamma_2 C - EE^T \gamma_1 C & \mathcal{A} - \gamma_2 C \end{bmatrix}_{6 \times 6}, E_z = \begin{bmatrix} E \\ E \end{bmatrix}_{6 \times 1}, \gamma_z = \begin{bmatrix} \gamma_1 \\ EE^T \gamma_1 \end{bmatrix}_{6 \times 1}. \quad (20)$$

4.1. Stability analysis

Following assumptions are made to illustrate the stability of CESO:

Assumption 1. Total disturbance f and its derivative $\frac{df(\cdot)}{dt} = h$ are bounded in the sense that $|f| \leq \mu_0$ and $|h| \leq \mu_1$ where $\mu_0, \mu_1 > 0$.

Assumption 2. Measurement noise (v) is bounded such that $|v| \leq \mu_2$ where $\mu_2 > 0$.

In order to highlight compromise in selection of observer bandwidth (ω_o) in terms of minimizing the effect of disturbance versus noise suppression, we apply the following change of variables on (19):

$$e_z \rightarrow \zeta_z := \mathbf{D}_z e_z, \quad (21)$$

where $\mathbf{D}_z := \text{diag}[1, \omega_o^{-1}, \omega_o^{-2}, 1, a^{-1}\omega_o^{-1}, a^{-2}\omega_o^{-2}]$, which results in

$$\begin{aligned} \dot{\zeta}_z &= \mathbf{D}_z \mathbf{A}_z \mathbf{D}_z^{-1} \zeta_z + \mathbf{D}_z \mathbf{E}_z h - \mathbf{D}_z \gamma_z v \\ &= \omega_o \mathbf{\Pi}_z \zeta_z + \omega_o^{-2} \bar{\mathbf{E}}_z h - \omega_o \bar{\gamma}_z v \end{aligned} \quad (22)$$

where $\text{eig}(\mathbf{\Pi}_z) = \{-1, -1, -1, -a, -a, -a\}$, $\bar{\mathbf{E}}_z = [\mathbf{E}_z^T, a^{-2}\mathbf{E}_z^T]^T$, $\bar{\gamma}_z = [\gamma_{1,1}, \gamma_{2,1}, \gamma_{3,1}, 0, 0, \gamma_{3,2}a]^T$.

Consider a Lyapunov function $V_z = \zeta_z^T \mathbf{P}_z \zeta_z$ where $\mathbf{P}_z$ is a symmetric positive definite matrix that satisfies $\mathbf{P}_z \mathbf{\Pi}_z + \mathbf{\Pi}_z^T \mathbf{P}_z = -\mathbf{I}$. Taking derivative of V_z along (22), results in

$$\dot{V}_z \leq -\omega_o \|\zeta_z\|^2 + 2\omega_o^{-2} \|\mathbf{P}_z\| \|\bar{\mathbf{E}}_z\| \|\zeta_z\| |h| + 2\omega_o \|\mathbf{P}_z\| \|\bar{\gamma}_z\| \|\zeta_z\| |v|, \quad (23)$$

which implies that

$$\lim_{t \rightarrow \infty} \|\zeta_z(t)\| \leq \kappa_1 \omega_o^{-3} \mu_1 + \kappa_2 \mu_2 \quad (24)$$

where κ_1, κ_2 are positive constants (Lakomy et al., 2022). Assuming $\omega_o \geq 1$, it can be shown that $a^{-2}\omega_o^{-2}\|e_z\| \leq \|\zeta_z\| \leq \|e_z\|$ from (21) which in turn yields the following bound for $\|e_z\|$:

$$\lim_{t \rightarrow \infty} \|e_z(t)\| \leq \kappa_1 a^2 \omega_o^{-1} \mu_1 + \kappa_2 a^2 \omega_o^2 \mu_2. \quad (25)$$

The actual estimation error vector $e := x - \hat{x} = [x_1 - \hat{x}_1, x_2 - \hat{x}_2, x_3 - \hat{x}_3]^T$, is a sub-vector of e_z , which implies $\|e\| \leq \|e_z\|$. Hence,

$$\lim_{t \rightarrow \infty} \|e(t)\| \leq \kappa_1 a^2 \omega_o^{-1} \mu_1 + \kappa_2 a^2 \omega_o^2 \mu_2. \quad (26)$$

The steady-state ultimate bound for $\|e\|$ illustrates that the effect of disturbance on estimation error reduces upon increasing the value of ω_o , however, the obtained estimates become more sensitive to the effect of measurement noise. It is also evident that practical convergence of estimation error is only possible in the neighbourhood of origin and that the estimation error cannot be made arbitrarily small by selecting higher values of ω_o .

Remark 1. It is to be noted that standard ESO used in conventional ADRC is a special case of CESO obtained for $N = 1$. Hence the design and stability analysis of conventional ESO has not been discussed separately. The stability bounds obtained for a two level CESO in (26) is also applicable for conventional ESO if we select $a = 1$.

5. Low-power higher-order extended state observer (LHESO)

In contrast to a standard ESO which assumes the disturbance to be constant (Ahmad and Ali, 2021b), a higher order ESO (HESO) (Godbole et al., 2012; Sira-Ramírez, 2018) incorporates a time polynomial type disturbance model into the augmented state-space form and is therefore more accurate in practical scenarios where disturbances are mostly time-varying. However, an HESO also suffers from the noise amplification problem that limits closed-loop performance in case of standard high-gain ESO. Furthermore, the numerical implementation issue (Khalil, 2017) that is encountered in standard high-gain ESO gets aggravated in case of HESO as the gains escalate to $\mathcal{O}(\omega_o^{n+m})$

where m denotes the number of state augmentations. To mitigate these issues, the HESO is implemented via a low-power construction introduced in Astolfi and Marconi (2015) which limits the magnitude of observer gain to $\mathcal{O}(\omega_o^2)$ and guarantees improved noise suppression in the estimates while simultaneously retaining all the good features of HESO such as asymptotic convergence of estimates in presence of disturbances that satisfy $\frac{d^m f(\cdot)}{dt^m} = 0$, when the effect of measurement noise is ignored.

In this study we consider a system having two state augmentations ($m = 2$) which is given by

$$\begin{aligned} \dot{x}^+ &= \tilde{\mathbf{A}} x^+ + b_0 \tilde{\mathbf{B}} u + \tilde{\mathbf{E}} g \\ y &= \tilde{\mathbf{C}} x^+ + v, \end{aligned} \quad (27)$$

where $x^+ = [x^T x_4]^T$, $x_4 = \dot{f} = g$, $\tilde{\mathbf{A}} = \begin{bmatrix} \mathbf{0} & \mathbf{I}_3 \\ \mathbf{0} & \mathbf{0} \end{bmatrix}_{4 \times 4}$, $\tilde{\mathbf{B}} = [0 \quad 1 \quad 0]^T_{1 \times 4}$, $\tilde{\mathbf{C}} = [1 \quad 0]_{1 \times 4}$ and $\tilde{\mathbf{E}} = [\mathbf{0} \quad 1]_{1 \times 4}^T$. The resulting LHESO designed for (27) is expressed using the following set of equations:

$$\begin{aligned} \dot{\hat{x}}_i &= \hat{x}_{i+1} + a_i b_0 u + \omega_o \beta_i (y_i - \hat{x}_i), \quad i = \{1, \dots, m+1\} \\ \dot{\hat{x}}_i &= \hat{x}_{i+1} + a_i b_0 u + \omega_o^2 \bar{\beta}_i (y_{i-1} - \hat{x}_{i-1}), \quad i = \{2, \dots, m\} \\ \dot{\hat{x}}_{m+1} &= \omega_o^2 \bar{\beta}_{m+1} (y_m - \hat{x}_m), \end{aligned} \quad (28)$$

where $y_i = \begin{cases} \hat{x}_i, & i = \{2, \dots, m\} \\ y, & i = 1, \end{cases}$ $a_i = \begin{cases} 1, & i = 2 \\ 0, & \text{otherwise.} \end{cases}$

Introducing estimation error vector as $e_\chi := x - \hat{x}$ where $\hat{x} := [\hat{x}_1^T, \hat{x}_2^T, \hat{x}_3^T]^T$, $\hat{x}_i := [\hat{x}_i, \hat{x}_{i+1}]^T$, $\chi = [x_1, x_2, x_2, x_3, x_3, x_4]^T$, we obtain error dynamics as

$$\dot{e}_\chi = \mathbf{A}_\chi e_\chi + \mathbf{H} g - \mathbf{F} v, \quad (29)$$

where $\mathbf{F} = [\beta_1 \omega_o, \bar{\beta}_1 \omega_o^2, \mathbf{0}_{1 \times 4}]^T$, $\mathbf{H} = [\mathbf{0}_{1 \times 5}, 1]^T$ while error matrix $\mathbf{A}_\chi = \Xi_3$ can be obtained recursively in the following manner:

$$\begin{aligned} \Xi_1 &= \mathbf{E}_1, \Xi_i = \begin{bmatrix} \Xi_{i-1} & \bar{\mathbf{N}}_i \\ \bar{\mathbf{Q}}_i & \mathbf{E}_i \end{bmatrix}, \mathbf{E}_i = \begin{bmatrix} -\beta_i \omega_o & 1 \\ -\bar{\beta}_i \omega_o^2 & 0 \end{bmatrix}, \bar{\mathbf{N}}_i = \begin{bmatrix} \mathbf{0}_{2(i-2) \times 2} \\ \mathbf{N} \end{bmatrix}, \\ \bar{\mathbf{Q}}_i &= [\mathbf{0}_{2 \times 2(i-2)} \quad \mathbf{Q}_i], \\ \mathbf{Q}_i &= \begin{bmatrix} 0 & \beta_i \omega_o \\ 0 & \bar{\beta}_i \omega_o^2 \end{bmatrix}, \mathbf{N} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, i = \{2, 3\}. \end{aligned} \quad (30)$$

The eigen values of $\mathbf{A}_\chi$ can be placed at $-\omega_o$ by selecting $\beta_i = 2$, $i = \{1, 2, 3\}$, $\bar{\beta}_1 = 3$, $\bar{\beta}_2 = 1$ and $\bar{\beta}_3 = \frac{1}{3}$, based on the concept of bandwidth parameterization proposed in Gao (2006). Such a selection also reduces the number of tuning parameters to unity and facilitates practical implementation. It is also worth noting that LHESO avoids the gain escalation problem which plagues high-gain observers (including ESO and CESO) and leads to numerical implementation complexity on fixed point digital signal processors by using gains that grow only up to ω_o^2 (Ahmad and Ali, 2021a; Khalil, 2017; Astolfi and Marconi, 2015).

5.1. Stability analysis

Similar to the case of CESO, the error vector e_χ is scaled in order to illustrate the effect of observer bandwidth on estimation error by using the transformation

$$\zeta := \mathbf{D} e_\chi, \quad (31)$$

where $\mathbf{D} := \text{blkdiag}(\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3)$ and $\mathbf{D}_i := \text{diag}(\omega_o^{-(i-1)}, \omega_o^{-i})$. Applying the preceding transformation on (29) gives

$$\begin{aligned} \dot{\zeta} &= \mathbf{D} \mathbf{A}_\chi \mathbf{D}^{-1} \zeta + \mathbf{D} \mathbf{H} g - \mathbf{D} \mathbf{F} v \\ &= \omega_o \mathbf{\Pi} \zeta + \omega_o^{-3} \mathbf{H} g - \omega_o \bar{\mathbf{F}} v, \end{aligned} \quad (32)$$

where $\text{eig}(\mathbf{\Pi}) = \{-1, -1, -1, -1, -1, -1\}$, and $\bar{\mathbf{F}} = [\beta_1, \bar{\beta}_1, \mathbf{0}_{1 \times 2m}]^T$. In order to show the boundedness of estimation error, we make the following assumption on the nature of total disturbance:

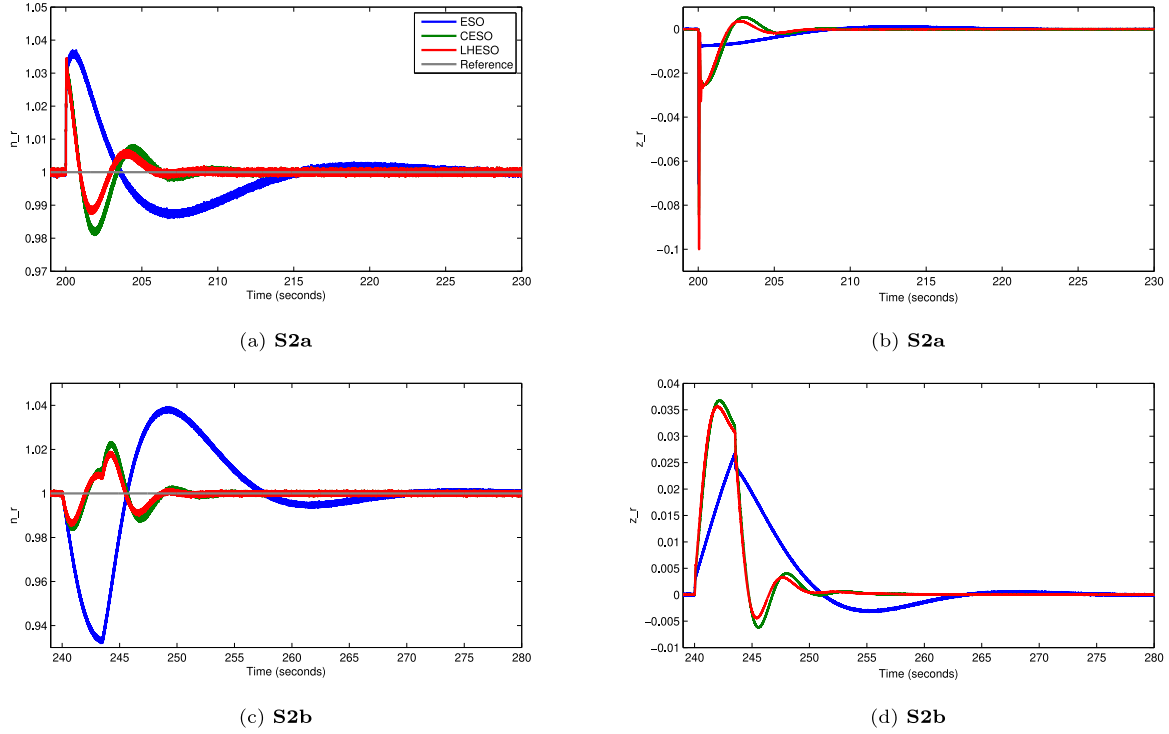


Fig. 2. Simulation results for Scenario 2 (S2a, S2b).

Assumption 3. Double derivative of total disturbance given by $\frac{d^2 f(\cdot)}{dt^2} = g$ is bounded in the sense that $|g| \leq \mu_3$ where $\mu_3 > 0$.

Remark 2. Although the time polynomial type disturbance model for $m = 2$ assumes $d^2 f/dt^2 = g = 0$, the exact nature of disturbance acting on the system is mostly unknown. Hence, a generalized assumption of g being bounded is more practical and is used in the present study to show the convergence of LHESO.

We define a Lyapunov function $V = \zeta^T P \zeta$ where P is a symmetric positive definite matrix that satisfies $P\bar{\Pi} + \bar{\Pi}^T P = -I$. Taking derivative of V along (32), results in

$$\dot{V} \leq -\omega_o \|\zeta\|^2 + 2\omega_o^{-3} \|P\| \|\zeta\| |g| + 2\omega_o \|P\bar{\Pi}\| \|\zeta\| |v|, \quad (33)$$

which implies that

$$\lim_{t \rightarrow \infty} \|\zeta(t)\| \leq \omega_o^{-4} \kappa_3 \mu_3 + \kappa_4 \mu_2, \quad (34)$$

where κ_3, κ_4 are positive constants (Ahmad and Ali, 2021a). Assuming $\omega_o \geq 1$ gives $\omega_o^{-3} \|e_\chi\| \leq \|\zeta\| \leq \|e_\chi\|$ from (31) which results in the following bound for $\|e_\chi\|$:

$$\lim_{t \rightarrow \infty} \|e_\chi(t)\| \leq \omega_o^{-1} \kappa_3 \mu_3 + \omega_o^3 \kappa_4 \mu_2. \quad (35)$$

Since the actual estimation error vector (e) is a sub-vector of e_χ , it can be written that $\|e\| \leq \|e_\chi\|$ which means

$$\lim_{t \rightarrow \infty} \|e(t)\| \leq \omega_o^{-1} \kappa_3 \mu_3 + \omega_o^3 \kappa_4 \mu_2. \quad (36)$$

Similar to the case of CESO, the preceding inequality illustrates the compromise in selection of observer bandwidth in terms of disturbance attenuation versus noise suppression. However, it is to be noted that the bounds obtained for both CESO and LHESO in (26) and (36) respectively, are conservative as they do not consider the frequency

content of the measurement noise signal. Consequently, the low-pass filtering capability of ESO's is also not evident from these expressions.

6. Analysis of noise suppression and disturbance attenuation

Noise suppression capabilities of ESO can be characterized using sinusoidal type measurement noise, similar to the approach highlighted in Astolfi and Marconi (2015), Astolfi et al. (2021). Analysing the asymptotic properties of estimation errors considering a composite sinusoidal noise signal of the form

$$v(t) = \sum_{i=1}^{n_v} s_i \sin(\epsilon^{-1} \omega_i t + \phi_i) \quad (37)$$

reveals that the relative degree between estimation error and measurement noise in case of two level CESO is two for e_1, e_2 and unity for total disturbance (e_3) (Ahmad and Ali, 2021a). This is due to the fact that the final estimates of x_1 and x_2 are obtained from level-two ESO which accepts output estimate from level-one ESO ($\hat{z}_{1,1}$) as a reference signal, whereas the estimate of x_3 (f) contains components from ESOs in each level i.e. $\hat{z}_{1,3}$ and $\hat{z}_{2,3}$. The higher relative degree contributes to noise suppression in the high frequency range and ensures better estimation quality compared to conventional ESO in case of e_1, e_2 whereas in case of e_3 , a lower observer bandwidth (ω_{o1}) for level-one ESO limits the noise content in $\hat{z}_{1,3}$. The argument can similarly be extended for N level CESO to show that the relative degree becomes N for e_1, e_2 but remains unity for e_3 . It is also evident that a lower relative degree in case of conventional ESO (obtained for $N = 1$) is the primary reason for poor noise suppression in the high-frequency range which in turn limits closed-loop performance in practical systems. On the other hand, the relative degree for LHESO is unity for e_1 , two for e_2 and three for e_3 i.e. it goes on increasing as we move from the output estimate towards disturbance estimate. This in turn ensures better noise suppression for LHESO as x_3 is most affected by the measurement noise due to the

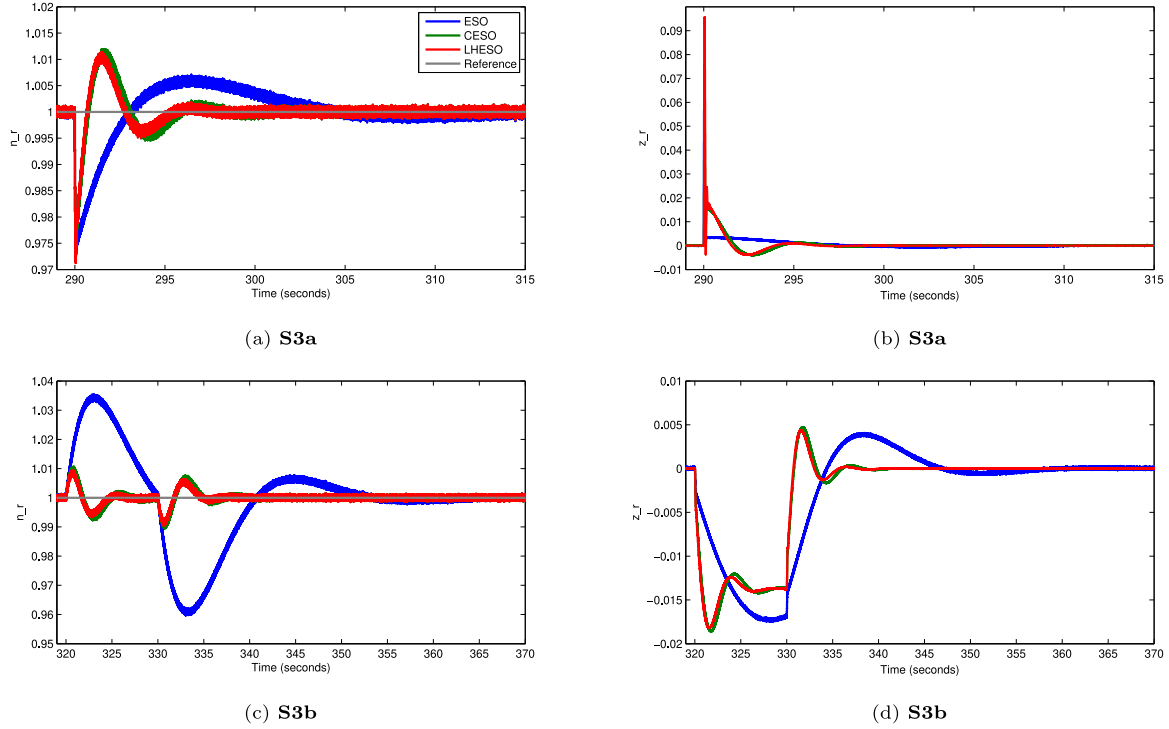


Fig. 3. Simulation results for Scenario 3 (S3a, S3b).

presence of double derivative term in its expression. However, unlike CESO where the relative degree increases for e_1, e_2 if we increase the number of cascade levels (N), the relative degree remains the same for LHESO irrespective of the choice of m as only the first three estimates are utilized in the control law.

As shown in Ahmad and Ali (2021a), virtual decomposition of total disturbance and its subsequent estimation via cascaded interconnection of ESOs results in accurate estimation of x_2 and x_3 even in case of ramp-type disturbance ($d^2f/dt^2 = 0$). This in turn yields improved disturbance attenuation compared to conventional ESO particularly in case of time-varying disturbances. However, the closed-loop system will exhibit a small steady-state error if the output estimate ($\hat{z}_{2,1}$) is used in the feedback control law in case of ramp-type disturbances due to the associated estimation error. In order to address this problem, the measured output signal (y) can directly be used in the feedback control law in place of its estimate at the cost of increased noise content in the control signal, as shown in Lakomy et al. (2022). Compared to CESO, LHESO ensures accurate estimation of all the system states for disturbances of type $d^m f/dt^m = 0$. This is also evident from the bounds obtained in (34) as $\lim_{t \rightarrow \infty} \|e(t)\| \rightarrow 0$ if $g = 0$. Hence LHESO yields error free reference tracking even in presence of ramp-type disturbances.

Remark 3: The main idea behind designing higher-order augmented observers with additional filtering is to leverage the enhanced estimation accuracy under time-varying disturbances while simultaneously suppressing the additional noise content introduced in the estimates via low-pass filtering. LHESO as well as CESO are two different approaches that exploit this very idea. As mentioned earlier, the higher state augmentation is not introduced by design in CESO but is inherently achieved via virtual decomposition followed by simultaneous estimation using interconnected ESOs. Furthermore, based on the results obtained in this paper as well as preliminary investigations in Ahmad et al. (2022), it can be expected that any combination of higher-order state augmentation (Sira-Ramírez, 2018; Godbole et al., 2012) along with additional filtering (Astolfi et al., 2021; Ahmad et al., 2023) will

result in significantly enhanced closed-loop performance compared to the standard ESO.

7. Simulation results

Numerical simulation for the closed loop system is performed in Simulink/MATLAB environment using ode3 solver with a fixed sampling time of 1 ms. Dynamics of PWR is simulated using the nonlinear model expressed in (1)–(8) with the system parameters defined as $\Lambda = 10^{-4}$ s, $\beta = 0.006019$, $\lambda = 0.15$ s $^{-1}$, $G_r = 0.0145\delta k/k$, $\mu_f = 26.3$ MW s/ $^\circ$ C, $T_{e0} = 290$ $^\circ$ C, $f_f = 0.92$, $P_0 = 2500$ MW (Liu et al., 2020). The effect of measurement noise is included by passing a Band-limited Gaussian white noise having power 10^{-11} , through an 8th-order Butterworth high-pass filter having pass band frequency of 200 rad/s. Performance of CESO and LHESO is compared to a conventional ESO to highlight the advantages of the two schemes. ESO, CESO and LHESO are implemented using the methods described in Sections 3–5 to obtain the estimates required for control law in (14), (15). The controller and observer parameters selected for the study are $\omega_c = 1$ rad/s and $\omega_o = \omega_{o2} = 40$ rad/s. The controller as well as observer bandwidths of all three ESOs (ω_{o2} in case of CESO) are kept same for a fair comparison. It is to be noted that the power level of PWR is regulated at the desired level using only a proportional-derivative type feedback control law as the integral action is inherently embedded into the feed-forward disturbance compensation implemented via (14).

Performance of the closed-loop system is evaluated for reference tracking as well as disturbance attenuation under nominal and perturbed scenarios considering three different scenarios defined as follows:

Scenario 1: Changes in power reference are applied to evaluate how quickly the power level of PWR reacts to such changes and attains the new reference under closed-loop control (Abdulraheem and Korolev, 2021; Liu et al., 2020).

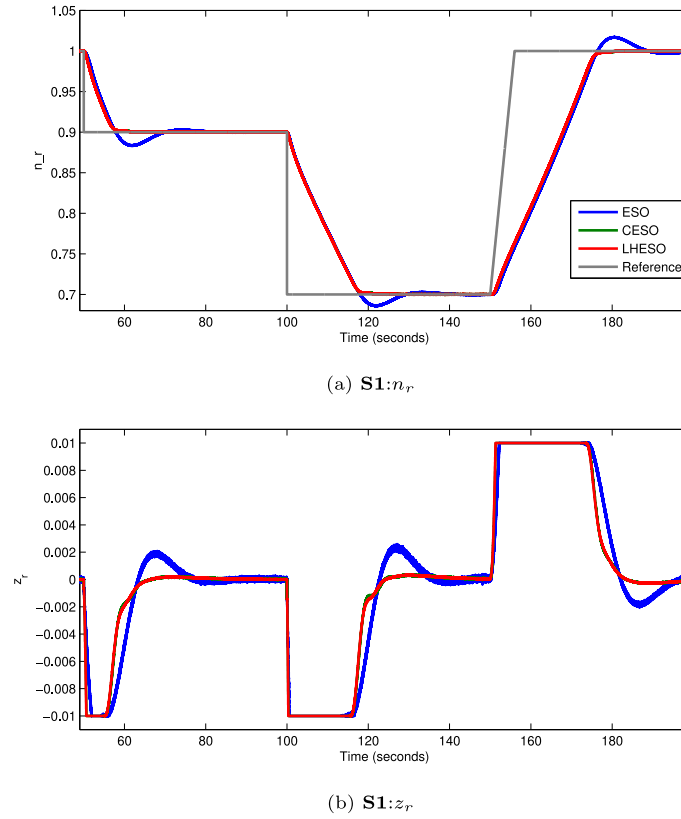


Fig. 4. Simulation results for Scenario 1 (S1a, S1b, S1c) under actuator constraints.

- **S1.a-** Negative 10% step change in power reference applied at $t = 50$ s.
- **S1.b-** Negative 20% step change in power reference applied at $t = 100$ s.
- **S1.c-** Positive 30% change in power reference applied at $t = 150$ s in ramp fashion having a slope of $0.05\%/s$.

Scenario 2: Abrupt changes in input temperature (T_e) is applied to evaluate the disturbance attenuation properties.

- **S2.a-** Step change of -2 °C in T_e applied at $t = 200$ s.
- **S2.b-** Gradual increase in T_e by 7 °C in ramp fashion having a slope of 2 °C/s, starting $t = 240$ s.

Scenario 3: Abrupt changes in reactivity (ρ) is applied to evaluate the disturbance attenuation properties (Dong et al., 2021).

- **S3.a-** Step decrease in reactivity by 0.0002 at $t = 290$ s.
- **S3.b-** Gradual increase in reactivity by 0.002 in ramp fashion with slope $0.0002/s$, starting at $t = 320$ s.

Nominal Operating Condition: Under nominal condition, the plots for test scenarios S1, S2 and S3 are obtained in Figs. 1, 2, and 3, respectively. For the reference tracking case in Fig. 1, it can be observed that LHESO and CESO based closed-loop system settles to the new power reference much faster that ESO based approach with no overshoot in case of LHESO and negligible overshoot for CESO. In order to quantify the performance gain obtained via noise suppressing ESOs, the *integral of absolute error* (IAE) criteria is computed as

$$IAE := \int_{t_i}^{t_f} |r(\tau) - y(\tau)| d\tau, \quad (38)$$

where t_i, t_f denote the initial and final time for respective scenarios while $|\cdot|$ computes the absolute value of the tracking error. The IAE values computed in Table 2 shows that the designed noise suppressing ESOs perform much better than conventional ESO (2.968) with LHESO

(1.208) having slightly better performance metrics than CESO (1.224). Furthermore, the control signal plot indicating velocity of the control rod (z_r) in Fig. 1(b) clearly shows in the zoomed-in view that LHESO based ADRC exhibits the best control quality (in terms of smoothness) and is least affected by the measurement noise present in the sensed data. Although CESO also suppresses the effect of measurement noise in control signal to a great extent (compared to conventional ESO), LHESO exhibits superior noise suppression due to its structural properties that result in increased relative degree. For abrupt changes in T_e , LHESO and CESO based schemes yield significantly better regulation compared to ESO, as shown in Section 2, with respective IAE values of 0.119, 0.152, 0.701, listed in Table 2. It is to be noted that the performance gap between conventional ESO and noise suppressing ESOs is more pronounced in case of ramp-type time-varying disturbances as shown in Fig. 2(c), due to the enhanced disturbance estimation capabilities of LHESO and CESO. For changes in ρ under S3, a similar trend can be observed in Fig. 3 and Table 2 for LHESO (0.072), CESO (0.090) and ESO (0.577) which highlights the performance gain obtained via designed ESOs.

Nominal Operating Condition with Actuator Limitation: Due to actuator limitations, the actual control rod speed (z_r) is a constrained value which needs to be taken into account while implementing the control algorithms. To illustrate the effect of this physical constraint, we consider that the control rod speed is limited in the range $z_r \in [-0.01, 0.01]$. In the simulation, this limitation can easily be implemented by introducing a saturation at the end of the control block where z_r is generated. However, it is necessary to incorporate the actuator limitation in the designed ESOs as well so that the estimators are able to accurately reconstruct the disturbance based on the actual value of z_r . Thus, in order to account for the actuator limitation, we saturate the control signal in the same range and then send the saturated z_r to the ESOs. The simulation results obtained for scenarios S1, S2 and S3 are shown in Figs. 4–6. It can be observed that despite the actuator

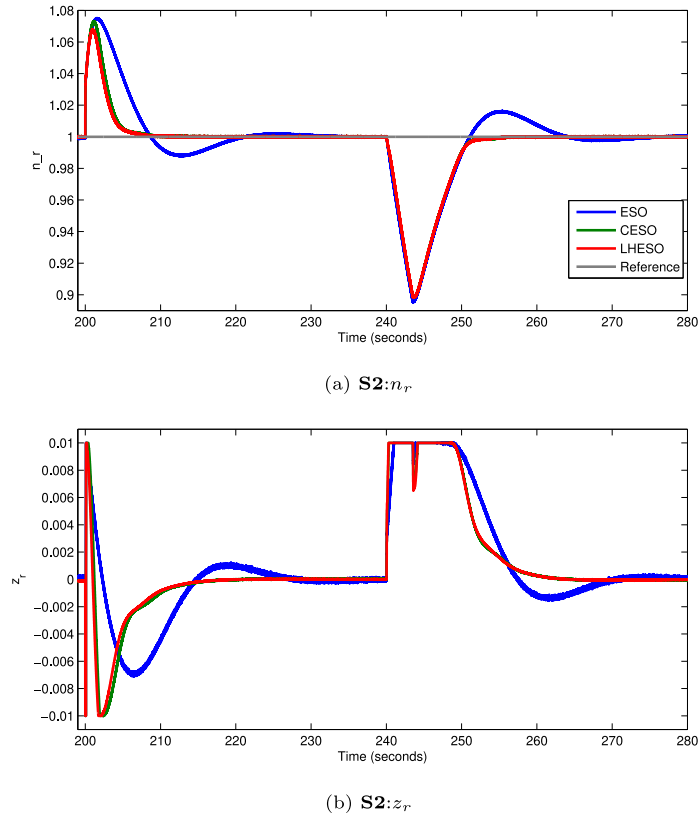


Fig. 5. Simulation results for Scenario 2 (S2a, S2b) under actuator constraints.

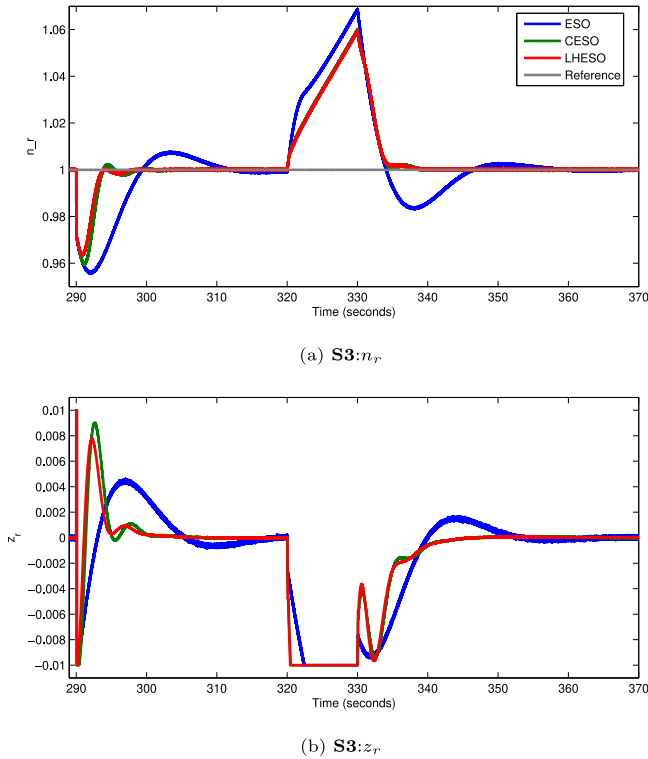


Fig. 6. Simulation results for Scenario 3 (S3a, S3b) under actuator constraints.

limitations, the designed noise suppressing ESOs result in significant performance improvement compared to conventional ESO based ADRC.

From the n_r plots in Figs. 4(a), 5(a) and 6(a), it can be noted that the response speed during the initial transient phase in case of reference tracking as well as disturbance regulation is almost the same for all three schemes as the control rod speed is saturated at its limit (evident in Figs. 4(b), 5(b) and 6(b)). Furthermore, it is also to be noted that LHESO and CESO based ADRC schemes exhibit faster recovery from the saturation region which in turn results in improved dynamic performance in terms of smaller overshoot/undershoot and faster settling to the desired equilibrium point compared to the conventional ADRC. The performance metrics in the form of IAE value obtained in case of actuator limitation is listed in Table 2. It can be noted that the IAE values of LHESO and CESO are much closer under actuator saturation, however, LHESO still performs slightly better.

Operation Under Parametric Perturbations: Robustness of the designed schemes is investigated by introducing perturbation into the PWR parameters G_r , λ and β where each system parameter is perturbed by $\pm 20\%$ giving rise to eight different perturbed scenarios (PS) in total which are defined as follows:

- PS1 := $\{G_r = +20\%, \lambda = +20\%, \beta = +20\%\}$
- PS2 := $\{G_r = +20\%, \lambda = +20\%, \beta = -20\%\}$
- PS3 := $\{G_r = +20\%, \lambda = -20\%, \beta = +20\%\}$
- PS4 := $\{G_r = +20\%, \lambda = -20\%, \beta = -20\%\}$
- PS5 := $\{G_r = -20\%, \lambda = +20\%, \beta = +20\%\}$
- PS6 := $\{G_r = -20\%, \lambda = +20\%, \beta = -20\%\}$
- PS7 := $\{G_r = -20\%, \lambda = -20\%, \beta = +20\%\}$
- PS8 := $\{G_r = -20\%, \lambda = -20\%, \beta = -20\%\}$

Plots obtained for the perturbed case are given in Fig. 7 which indicate that the designed schemes are robust towards large perturbations in system parameters and are able to ensure system stability while tracking reference changes as well as under the effect of variations in T_e and reactivity. The corresponding IAE values for all three scenarios are listed under Table 3. The spread of the plots in time domain (Fig. 7) and

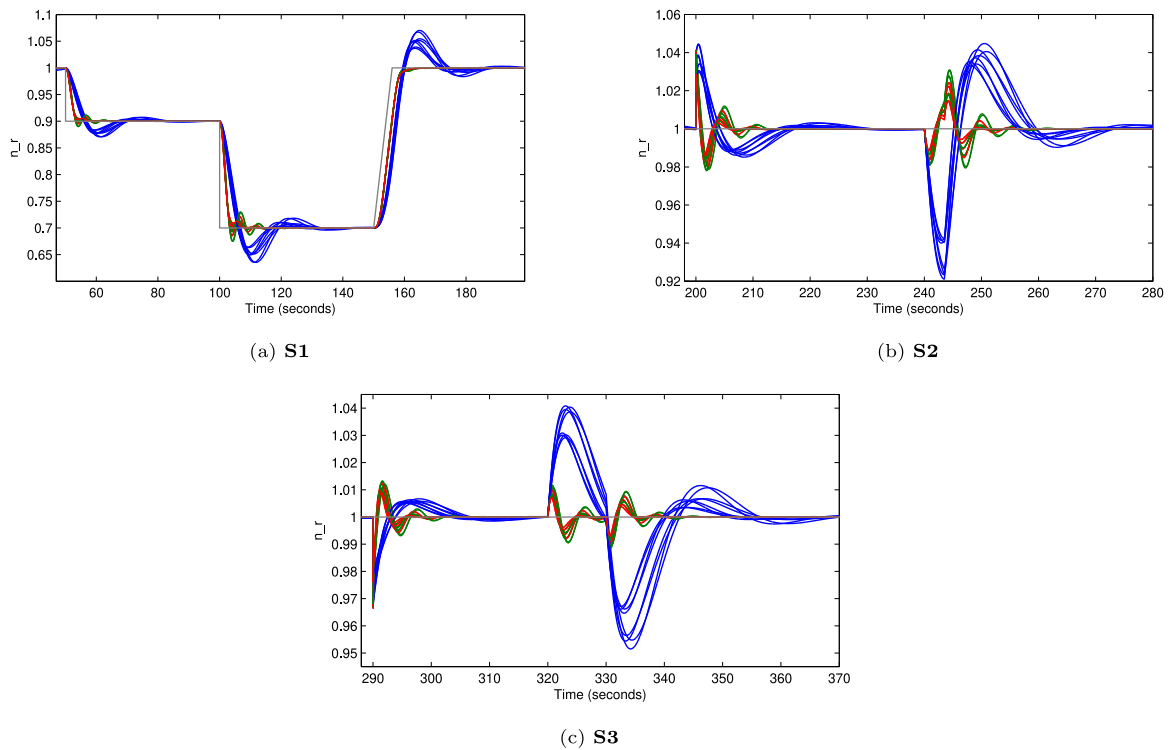


Fig. 7. Closed-loop system response under parametric perturbations without the effect of measurement noise (legend: red-LHESO, green-CESO, blue-ESO, grey-Reference).

Table 2

IAE values for ESO, CESO and LHESO based ADRC schemes under nominal condition.

	W/o actuator limit			With actuator limit		
	S1	S2	S3	S1	S2	S3
ESO	2.968	0.701	0.577	5.722	1.162	0.991
CESO	1.224	0.152	0.090	5.156	0.757	0.570
LHESO	1.208	0.119	0.072	5.142	0.732	0.545

variation of the performance metrics (Table 3) is an indicator of the extent to which parametric perturbations affect closed-loop performance. It is evident from the time domain plots and performance metrics that LHESO and CESO are more robust towards parametric perturbations as their closed-loop performance is less affected compared to conventional ESO based ADRC. It is also to be noted that the performance of LHESO and CESO based ADRC is better than conventional ESO based ADRC under all combinations of parametric perturbations.

A comparison of ESO, CESO and LHESO designed for an n th order system is presented in Table 4 to highlight their distinguishing features.

8. Conclusion

This paper investigates the efficiency of recently proposed noise suppressing ESOs for control of PWR under practical operating conditions. Convergence analysis of level two CESO and LHESO with two state augmentations are presented using Lyapunov analysis considering the effect of total disturbance as well as sensor noise in the measured signal. Through extensive simulation studies, it is shown that the noise suppressing ESOs yield improved closed-loop performance compared to conventional ESO in terms of reference tracking as well as attenuation of step and ramp-type disturbances. A similar trend is observed even when the actuator constraints are taken into consideration where LHESO and CESO exhibit faster recovery from the saturation region compared to standard ESO resulting in significantly improved closed-loop performance particularly in case of disturbance attenuation. In addition to nominal operating condition, the closed-loop performance

Table 3

IAE values for ESO, CESO and LHESO based ADRC schemes under perturbed condition.

		ESO	CESO	LHESO
PS1	S1	2.775	1.222	1.208
	S2	0.624	0.131	0.102
	S3	0.490	0.078	0.062
PS2	S1	2.398	1.207	1.206
	S2	0.577	0.112	0.124
	S3	0.435	0.067	0.055
PS3	S1	2.925	1.223	1.208
	S2	0.606	0.128	0.100
	S3	0.489	0.077	0.061
PS4	S1	2.516	1.207	1.207
	S2	0.569	0.111	0.124
	S3	0.438	0.067	0.054
PS5	S1	3.699	1.382	1.283
	S2	0.913	0.249	0.184
	S3	0.813	0.142	0.108
PS6	S1	3.076	1.225	1.207
	S2	0.825	0.185	0.167
	S3	0.701	0.108	0.079
PS7	S1	3.927	1.382	1.285
	S2	0.891	0.238	0.178
	S3	0.804	0.136	0.104
PS8	S1	3.258	1.229	1.207
	S2	0.815	0.182	0.166
	S3	0.705	0.106	0.079

is evaluated under eight different combinations of parametric perturbations to evaluate the robustness of the three ESOs. Structural properties of CESO and LHESO are also discussed to highlight their advantages as well as limitations. Further improvements in performance as well as noise suppression can be obtained by combining the two structures and will be investigated in future works. Further validation of the proposed

Table 4
Comparison of different observer structures for an n th-order system.

	ESO	CESO	LHESO
Design parameters	ω_c, ω_o	$\omega_c, \omega_{oi}(i=1 \text{ to } N), N$	ω_c, ω_o, m
Observer tuning	$\gamma_i = \frac{(n+1)!}{i!(n+1-i)!} \omega_o^i, i=1 \text{ to } n+1$	$\gamma_{j,i} = \frac{(n+1)!}{i!(n+1-i)!} \omega_o^i, i=1 \text{ to } n+1, j=1 \text{ to } N$	$\beta_i = 2, \tilde{\beta}_i = \frac{(n+m-i)}{i}, i=1 \text{ to } n+m-1$
Observer states	$n+1$	$N(n+1)$	$2(n+m-1)$
Assumption for convergence	$\dot{f}$ is bounded	$\dot{f}$ is bounded	$f^{(m)}$ is bounded
Maximum gain	$\mathcal{O}(\omega_o^{n+1})$	$\mathcal{O}(\omega_o^{n+1})$	$\mathcal{O}(\omega_o^n)$
Accuracy in presence of time-varying f , $f^{(m)} = 0, (m = N)$	All state estimates are inaccurate	At least $\dot{f}$ is accurate	All state estimates are accurate
Relative degree between e_i and v	Unity for all i	N for $i = \{1, \dots, n\}$ and 1 for $i = n+1$	i

strategies using a more detailed and realistic nuclear reactor model will be considered in a future work.

CRedit authorship contribution statement

Saif Ahmad: Conceptualization, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Kamal Kayode Abdulraheem:** Methodology, Writing – original draft, Writing – review & editing. **Andrei Olegovich Tolokonsky:** Supervision, Writing – review & editing. **Hafiz Ahmed:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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